TITLE

Research Proposal

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Declaration

I, Sashin Samlal hereby declare the contents of this research document to be my own work. This document is submitted for degree purposes in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.
Date: May 25, 2026

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Chapter 1

Introduction

The barrier to entry for the RoboCup Small Size League (SSL) is primarily defined by the prohibitive cost of specialized robotic hardware and the inherent complexity of high-level algorithm implementation. For instance, a research team must purchase specialized omnidirectional wheels and high-torque brushless motors, costing thousands of dollars, to build a single competitive unit. These limitations often exclude researchers and institutions with limited funding from contributing to the advancement of multi-agent systems and competitive robotics.

In a South African context, where the top 10% of the population captures approximately 70% of the national income [Czajka and Gethin 2026], this creates a severe digital divide that hinders the potential growth of the robotics field. Because robotics education is highly resource-intensive and relies on expensive hardware, software, and stable internet infrastructure under-resourced rural public schools are frequently excluded from these advancements [Mapheto 2023]. Consequently, this financial barrier restricts the broader societal impact of robotics innovation, confining the development of vital 21st-century technological skills primarily to well-resourced private institutions and a privileged economic minority.

A critical solution to this cost barrier lies in the development of a standardized, competition-capable robotic platform that utilizes additive manufacturing and open-source middleware to simplify hardware-software integration. Such a design provides an accessible entry point, fostering wider participation in the SSL and stimulating broader interest in autonomous systems.

Historically, low-cost approaches have relied on a "hobbyist" paradigm, utilizing off-the-shelf boards and structural components sourced from general hardware suppliers, such as sheet metal. While various open-source middleware solutions exist to facilitate software standardization, integrating disparate hobbyist components remains a significant challenge due to proprietary communication protocols and manufacturer-specific constraints. Beyond component integration, traditional manufacturing methods like welding or metalwork require specialized skills to meet the strict dimensional tolerances mandated by league regulations.

However, the rapid advancement and cost-effectiveness of 3D printing technology now allow for the production of standardized, modular, and easily reproducible structural components. By integrating this hardware with the Robot Operating System (ROS) middleware, it is possible to create a standardized interface that abstracts

low-level control and enables seamless transition between simulation and physical deployment.

The establishment of this unified hardware-software framework further permits the implementation of standardized benchmarking suites, providing a rigorous environment for the performance evaluation of diverse robotic components and autonomous strategies.



Chapter 2

Background

The RoboCup SSL represents a highly constrained environment designed to promote research in fast-paced multi-robot systems. To understand the benchmarking of a custom low-cost platform against standard solutions, it is necessary to establish the regulatory and physical parameters of the league. Table 2.1 summarizes the core technical standards mandated by the league.

Table 2.1: RoboCup SSL Division B Technical Standards [RoboCup Small Size League Technical Committee 2026].

Parameter	Regulation	Design Impact
Robot Volume	$180\,\mathrm{mm} \times 150\,\mathrm{mm}$ height	Constrains motor size and wheel geometry.
Ball Velocity	Max $6.5\,\mathrm{m/second}$	Defines limits for solenoid/pneumatic kicker systems.
Ball Interaction	$< 20\%$ coverage	Prevents ball hoarding; requires precise dribbler geometry.
Halt Latency	< 2.0s to zero velocity	Mandates high-frequency braking control.
Stop Speed	Max $1.5\,\mathrm{m/second}$	Requires real-time speed regulation in the control stack.
Vision	Overhead SSL-Vision	Necessitates dot-pattern ID and distributed processing.

2.1 Design Implications of Technical Constraints

These regulations are not merely boundary conditions; they directly influence the scientific and engineering trade-offs required to build a competition-capable platform.

2.1.1 Spatial Constraints and Mechanical Packaging

The strict $180\text{mm} \times 150\text{mm}$ cylindrical volume creates a complex "packing problem" for the robotic chassis. In competitive, high-cost units, this volume is typically maximized using four-wheel omnidirectional drives to achieve holonomic movement. However, implementing such a drive requires precise geometric placement of four independent motors and wheels within the circular convex hull, significantly increasing mechanical complexity and decreasing the space available for batteries and low-level electronics [Avidano *et al.* 2023].

By contrast, a differential-drive system simplifies this spatial constraint by utilizing only two independent drive wheels. While this introduces non-holonomic constraints, which require the robot to rotate before translating, it allows for a more stable and modular internal layout. This study benchmarks whether such a simplified mechanical design, optimized for 3D printing, can still meet the maneuverability requirements of the league while remaining financially accessible.

To model the motion of this differential-drive configuration, we establish the kinematic equations relating the wheel velocities to the robot velocities. Let r represent the wheel radius, and let L denote the distance between the two drive wheels. The state of the robot in the global coordinate frame is defined by the vector $\mathbf{x} = [x, y, \theta]^T$, where (x, y) represents the midpoint of the wheel axle and θ is the heading angle. The linear velocity v and angular velocity ω in the robot's local frame are calculated from the right wheel angular velocity $\dot{\phi}_R$ and left wheel angular velocity $\dot{\phi}_L$:

$$v = \frac{r}{2}(\dot{\phi}_R + \dot{\phi}_L) \tag{2.1}$$

$$\omega = \frac{r}{L}(\dot{\phi}_R - \dot{\phi}_L) \tag{2.2}$$

The global state update equations are:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \tag{2.3}$$

2.1.2 Temporal Constraints and Control Theory

The league's safety protocols, specifically the "Halt" and "Stop" commands, impose rigorous temporal constraints on the control architecture. The requirement to reach zero velocity within two seconds of a "Halt" signal necessitates a control stack capable of high-frequency sampling and rapid deceleration profiles. In addition to halting quickly, maintaining a maximum speed of 1.5m/s during "Stop" phases requires precise velocity estimation. These constraints motivate the integration of a Proportional-Integral-Derivative (PID) or Model Predictive Control (MPC) loop within the software stack to manage discretization delays and sensor noise, ensuring that the physical platform adheres to the safety standards defined by the referee box (refbox) [Pacheco and Luo 2015].

The velocity controller regulates the wheel angular velocities. The error $e(t)$ between the target wheel velocity $\dot{\phi}_{target}(t)$ and measured velocity $\dot{\phi}_{measured}(t)$ is:

$$e(t) = \dot{\phi}_{target}(t) - \dot{\phi}_{measured}(t) \tag{2.4}$$

The control signal $u(t)$ is calculated using a continuous PID control law:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{2.5}$$

where K_p, K_i, and K_d represent the proportional, integral, and derivative gains respectively. For digital implementation on the microcontroller, this control law is discretized at a sampling interval Δt:

$$u_k = K_p e_k + K_i \Delta t \sum_{j=0}^{k} e_j + \frac{K_d}{\Delta t}(e_k - e_{k-1}) \tag{2.6}$$

where k represents the current time step.

2.1.3 Information Constraints and Distributed Systems

The prohibition of human intervention and the reliance on the centralized SSL-Vision system motivate a distributed "software glue" architecture. Since all positional data is generated by overhead cameras and distributed via wireless packets, the robot cannot rely on local on-board vision.

This creates a dependency on network latency and external state estimation. To address this, the proposed architecture utilizes a Single Board Computer (SBC) running ROS to bridge the gap between centralized vision commands and local motor execution [Smit 2011; Joseph 2018]. This distributed intelligence allows the platform to handle global strategy while maintaining local control stability, an essential requirement for high-speed autonomous play.

The centralized vision server transmits tracking data over a network via User Datagram Protocol (UDP) multicast packets. The payload of each packet is serialized using Google Protocol Buffers, containing a nested data structure defined as:

```
message SSL_DetectionFrame {
  required double t_capture = 1;
  required double t_sent = 2;
  required uint32 camera_id = 3;
  repeated SSL_DetectionBall balls = 4;
  repeated SSL_DetectionRobot robots_yellow = 5;
  repeated SSL_DetectionRobot robots_blue = 6;
}
```

where each robot packet contains its team ID, coordinates (x, y), and orientation θ.

Chapter 3

Related Work

The development of robotic platforms for competitive environments like the RoboCup SSL has evolved through distinct shifts in locomotion strategy, software architecture, and manufacturing philosophy. This section explores these transitions to motivate the design of a low-cost, 3D-printed benchmarking platform.

3.0.1 Locomotion and Drive Mechanisms

The high-speed, dynamic nature of robotic soccer has traditionally necessitated the use of complex holonomic drives to allow for simultaneous translation and rotation. While platforms designed for the SSL overwhelmingly utilize omnidirectional drive mechanisms, their dominance is justified by the essential requirement to decouple heading from velocity vectors. This decoupling allows a robot to maintain its orientation toward the ball or a target while translating in any direction, a capability that is essential for high-frequency ball interception and defensive "side-shuffling."

Despite their tactical superiority, omnidirectional systems impose a significant "complexity tax." Implementing a four-wheel omni-drive requires the geometric synchronization of four independent motors, which increases the probability of mechanical failure and reduces the internal volume available for power-dense components. Recent literature, such as the *First Order* Team Description Paper (TDP) [Liang *et al.* 2025], highlights that the transition to a compact holonomic base often requires a complete mechanical redesign to manage these spatial trade-offs.

By contrast, the differential-drive approach represents a strategic sacrifice of agility for the sake of mechanical durability and cost-efficiency. The non-holonomic penalty represents a key performance bottleneck; differential robots must perform a "rotate-then-translate" maneuver, which introduces a measurable latency in response to rapid changes in ball trajectory. Tactically, this limitation makes differential platforms less effective in goal-tending scenarios where lateral movement is paramount.

Quantitatively, benchmarking studies demonstrate that differential drive robots can experience an increase in "Time-to-Target" metrics and path planning complexity compared to holonomic peers [Chavan and Minase 2015]. However, for a low-cost 3D-printed platform, the differential drive simplifies the control stack and reduces manufacturing barriers. The proposed project utilizes the differential approach, following the success of the *MICROMVP* platform [Yu *et al.* 2017], to determine if the loss in agility

is a justifiable trade-off for a benchmarking-focused research platform.

3.0.2 Evolution of System Architecture and Middleware

The development of the RoboCup SSL has been marked by a fundamental shift from centralized "dumb" agents to distributed, intelligent architectures. Early systems relied on a single off-field computer to process vision data and transmit low-level actuator duty cycles directly to robots. However, the increasing complexity of multi-robot coordination and the need to reduce dependency on high-bandwidth wireless links motivated the adoption of on-board SBCs capable of local state estimation and motor control [Smit 2011].

The transition to distributed intelligence necessitated a standardized "software glue" to manage inter-process communication. While early research utilized the Player/Stage middleware, ROS emerged as the dominant standard due to its modular "computational graph" and extensive ecosystem of visualization and diagnostic tools [Joseph 2018]. ROS simplified the integration of disparate hardware by providing a common messaging interface, allowing researchers to focus on high-level algorithm development rather than low-level driver implementation.

Despite its success, the evolution from ROS1 to ROS2 represents a significant technical milestone for the SSL environment. ROS1 utilized a centralized "ROS Master" and a TCP-based transport layer, which introduced a single point of failure and was ill-suited for the unreliable wireless networks of a competition floor. In contrast, ROS2 adopts a decentralized architecture based on the Data Distribution Service (DDS). By eliminating the Master and introducing configurable Quality of Service (QoS) policies (such as "Best Effort" and "Transient Local"), ROS2 provides the reliability and multi-robot discovery capabilities required for dynamic agent interaction [Macenski *et al.* 2022].

However, the adoption of ROS introduces significant performance trade-offs, particularly regarding real-time determinism. Standard ROS running on a non-preemptive Linux kernel is inherently non-deterministic; the standard Completely Fair Scheduler (CFS) prioritizes overall throughput over meeting strict millisecond deadlines [Kronauer *et al.* 2021]. This introduces "jitter" into control loops that can degrade the stability of high-speed platforms. In addition to scheduling jitter, the communication overhead of message serialization and deserialization, the process of transforming data structures into binary buffers for the DDS, can introduce latencies that impact the 60Hz feedback cycles provided by SSL-Vision [Gutiérrez *et al.* 2018]. This study analyzes whether these architectural overheads can be mitigated through zero-copy intra-process communication or if they necessitate the use of PREEMPT_RT kernel patches to bridge the "reality gap" in software timing.

3.0.3 Accessibility through Cost Optimization

The high barrier to entry in robotics is often defined by manufacturing complexity. To overcome this, recent literature highlights a move away from expensive machined metal toward additive manufacturing. Teams increasingly utilize 3D-printed plastics to reduce costs and increase design flexibility for complex components [Liang *et al.* 2025].

This accessibility is further enhanced by combining 3D printing with off-the-shelf electronics. The *MICROMVP* platform demonstrates that a highly affordable micro-scale testbed is achievable through this combination [Yu *et al.* 2017]. This layout is extended by Joseph [2018], who provide a framework for bridging high-level ROS nodes running on an SBC directly to low-level, inexpensive microcontrollers, an approach central to this project's hardware design.

3.0.4 Bridging the Gap: Simulation vs. Physical Validation

Validation of multi-agent algorithms often occurs in virtual environments to bypass the costs of physical testing. For the RoboCup Middle Size League, Yao *et al.* [2015] develop a simulation system based on ROS and Gazebo, utilizing a custom model plugin to mathematically simulate locomotion.

While such simulations simplify robots into basic cylindrical collision elements to preserve computational speed, they require the injection of Gaussian noise to mimic real-world sensor inaccuracies [Yao *et al.* 2015]. This "reality gap" motivates the necessity of the current project: by building a physical benchmarking platform, we can quantitatively evaluate how algorithms perform against real-world friction, battery dynamics, and discretization delays that simplified virtual models often fail to capture within the constraints defined by the official league rules [RoboCup Small Size League Technical Committee 2026].

Chapter 4

Methodology

4.1 Hardware Architecture and Component Selection

The physical chassis of the robotic platform adopts a cylindrical geometry to facilitate rotation within confined spatial envelopes, drawing structural design principles from the multi-vehicle platform presented by Yu *et al.* [2017]. Integrating three distinct cutouts, the base plate features two arch-shaped apertures accommodating the primary differential wheels alongside a third rectangular opening for a passive caster wheel that maintains vertical stability. Structural fabrication utilizes three-dimensional (3D) printing techniques to support rapid prototyping and low-cost replication. Visual representations of the three-dimensional CAD chassis models are detailed in Appendix A, showing the top view (Figure A.1a), front view (Figure A.1b), and side view (Figure A.1c). An open-top design characterizes the casing, which employs thickened mounting structures to secure the microcontroller. Convective airflow promoted by this open configuration cools internal electronics and provides direct physical access to the motor terminals. Soldered connections are applied to all electrical interfaces to maximize conductivity and minimize the probability of short-circuits during high-acceleration maneuvers.

The physical dimensions of the platform are constrained by the RoboCup SSL regulations, which mandate that all participating robots must fit within a cylinder of 180 mm diameter and 150 mm height [RoboCup Small Size League Technical Committee 2026]. Remaining comfortably within these limits, the proposed design utilizes a footprint of 100 mm diameter and 100 mm height. Symmetrical cylinders minimize rotational inertia and simplify the 3D-printing process relative to asymmetric geometries. Internal space accommodates a Raspberry Pi Pico microcontroller board, which measures 51 mm by 21 mm. A visual layout of the microcontroller is detailed in Appendix B (Figure B.1), which is retrieved from the official product specification sheet [Raspberry Pi Foundation 2026]. Running micro-ROS, this board executes a specialized client library enabling microcontrollers to communicate directly with a central ROS environment [Macenski *et al.* 2022]. Low unit cost and compatibility with hardware extension boards, such as Wi-Fi and camera modules, support modular hardware scaling. Although an onboard camera is excluded from the initial prototype to minimize cost, the physical chassis design reserves mounting space to accommodate future computer vision integration.

Locomotion is driven by two spur gear DC motors, representing a mechatronic configuration modeled after the motor drive systems that Salem [2013] analyzes. Spur gear motors are selected due to their affordability, general availability, and integrated internal gearheads, which increase output torque at low rotational velocities. To track and regulate the dynamic motion of the platform in real time, the chassis houses an MPU6050 three-axis accelerometer. This sensor communicates with the microcontroller via an Inter-Integrated Circuit (I^2C) interface and fits within a compact spatial footprint, facilitating mounting near the center of mass of the robot to collect accurate acceleration data.

4.2 Distributed Software Stack and Communication

The control software is structured around ROS and its microcontroller-compatible extension, micro-ROS, which establishes a distributed publisher-subscriber communication network [Macenski *et al.* 2022]. Primary computational operations, including navigation and path-planning algorithms, run on a centralized laptop serving as the ROS master node. Velocity vectors calculated by this master node are published as movement commands to specific command topics. Subscribing to these topics, the micro-ROS client running on the Raspberry Pi Pico translates the velocity commands into motor control signals for execution. Closing the feedback loop, the microcontroller publishes local acceleration data from the MPU6050 accelerometer back to the ROS master node. Sensor streams received by the master node are integrated into the central controller to dynamically adjust subsequent velocity commands.

Position and orientation tracking are achieved via an external overhead camera system. Overhead detection relies on a unique visual marker, or sigil, affixed to the top of the robot chassis, which allows the tracking software to resolve the coordinate position and heading angle of the vehicle in real time. This localization framework follows the multi-vehicle tracking pipeline that Yu *et al.* [2017] describe. Coordinate streams are processed by the central laptop to determine the position of the robot relative to the SSL playing field, serving as the input for path-generation algorithms. Figure 4.1 illustrates this distributed system architecture and the data pipeline mapping raw camera coordinates and accelerometer feedback to wheel velocities.

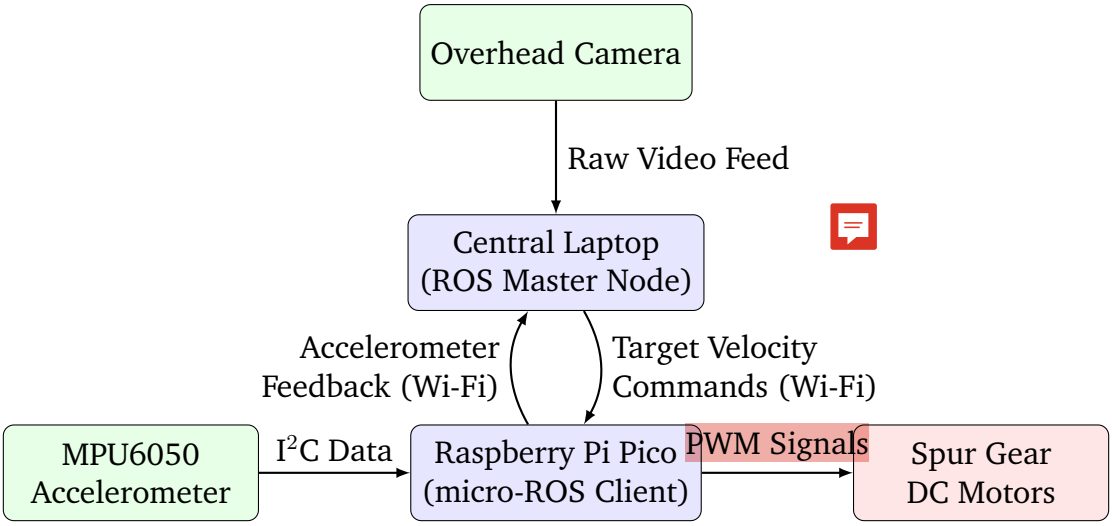


Figure 4.1: Distributed system architecture and data pipeline.

To regulate motor speeds and minimize tracking errors, the microcontroller implements a PID control algorithm. Control input $u(t)$ sent to the motor drivers is defined as:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{4.1}$$

where $e(t) = v_{\text{target}}(t) - v_{\text{measured}}(t)$ is the difference between the target velocity commanded by the ROS master and the actual velocity measured by the MPU6050 accelerometer. Experimental tuning determines the PID parameters K_p, K_i, and K_d to prevent overshoot and maintain controller stability under rapid deceleration profiles. Selection of the PID algorithm is justified by its low computational overhead, which fits within the processing constraints of the Raspberry Pi Pico microcontroller, and its established effectiveness in mobile robot speed regulation, which Pacheco and Luo [2015] demonstrate.

4.3 Benchmarking Protocols and Experimental Setup

Performance evaluation is conducted by comparing the manufactured differential-drive robot against a standard laboratory TurtleBot. Baseline parameters are established by subjecting the TurtleBot to an identical battery of tests, defining the target benchmarks for the proposed platform. If the TurtleBot fails to meet the operational standards defined by the RoboCup SSL rules, the official league specifications serve as the primary benchmarking reference [RoboCup Small Size League Technical Committee 2026].

The first benchmark evaluates the maximum velocity sustained by the vehicle, denoted as v_{max}. Calculation of this value utilizes the maximum change in displacement over time:

$$v_{\text{max}} = \max \frac{\Delta d}{\Delta t} \tag{4.2}$$

measured over a straight-line trajectory of 5 meters. This metric is necessary because RoboCup SSL rules restrict maximum robot velocity to 1.5 m/s to ensure safety and

prevent high-speed collisions on the field [RoboCup Small Size League Technical Committee 2026].

The second benchmark measures the stopping time, t_{stop}, which evaluates the efficiency of the deceleration control loop. Stopping time is defined as the elapsed time between the transmission of a halt command, t_{halt}, and the moment the robot's velocity reaches zero, t_{zero}:

$$t_{\text{stop}} = t_{\text{zero}} - t_{\text{halt}} \tag{4.3}$$

Compliance with the SSL regulations is determined by this metric, which mandates a maximum stopping grace period of 2.0 seconds after a halt command is broadcast to maintain game control [RoboCup Small Size League Technical Committee 2026].

The third benchmark assesses the trajectory accuracy of the platform using an M-shaped path test. Path fidelity, E_{path}, is quantified using the root-mean-square error (RMSE) between the target path coordinates and the actual coordinates tracked by the overhead camera:

$$E_{\text{path}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2]} \tag{4.4}$$

where (x_i, y_i) represents the target coordinate pair at step i, and $(\hat{x}_i, \hat{y}_i)$ represents the coordinates recorded by the vision system. Tracking performance evaluates the ability of the non-holonomic differential-drive platform to execute sharp turns and maintain precise trajectory tracking under tight spatial constraints.

The fourth benchmark measures command response delay, or latency, denoted as τ_{latency}. Latency computation uses the difference:

$$\tau_{\text{latency}} = t_{\text{motion}} - t_{\text{command}} \tag{4.5}$$

where t_{command} is the exact timestamp when a motion command is sent from the ROS master, and t_{motion} is the timestamp when the onboard accelerometer registers physical movement exceeding a baseline threshold. Quantifying this delay is necessary to verify the real-time communication performance of the ROS and micro-ROS message bridge under active play conditions [Kronauer *et al.* 2021].

The fifth benchmark measures the stopping distance, d_{stop}, which is the physical displacement traveled after a deceleration command is issued. Euclidean distance is used to calculate the stopping distance between the position of the robot when the halt command is issued and its final resting position:

$$d_{\text{stop}} = \|\mathbf{p}(t_{\text{halt}}) - \mathbf{p}(t_{\text{zero}})\| \tag{4.6}$$

where $\mathbf{p}(t)$ represents the two-dimensional position vector of the robot. Characterizing stopping distance is required to evaluate the safety of the deceleration control profile and ensure the vehicle can avoid obstacles and field boundaries when halted.

The sixth benchmark determines the turning radius, R_{turn}, to establish the minimum spatial envelope required for direction changes. Measurement of this metric requires a physical marker to be attached to the outermost edge of the robot chassis so that it contacts the ground. Rotation of the robot is commanded through a 180-degree angle, and the radius of the resulting arc drawn on the surface is measured. Establishing this

turning radius allows the central path-planning algorithms to calculate valid trajectories that avoid collisions with other robots on the playing field.

The final benchmark measures system consistency by computing the statistical dispersion of the collected metrics. Statistical analysis involves executing 10 iterations of each test and calculating the standard deviation, σ_x, for each performance metric:

$$\sigma_x = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}(x_j - \bar{x})^2} \tag{4.7}$$

where x_j is the metric value recorded in iteration j, $\bar{x}$ is the mean value across all iterations, and $n = 10$. Evaluating this dispersion is necessary to determine the reproducibility of the robot's physical performance, ensuring that the control algorithms can rely on predictable mechanical responses.

Chapter 5

Workplan

Table 5.1: Proposed Schedule of Work (June 30 – August 30)

Task	Description	Start Date	End Date
1	Chassis fabrication: 3D printing of chassis, base plate, motor brackets, and casing.	30-06-2026	04-07-2026
2	Mechanical assembly: Mount Raspberry Pi Pico, MPU6050, motors, caster, and solder connections.	05-07-2026	08-07-2026
3	Firmware development: Implement Pico motor control, I^2C driver, and sensor reading routines.	09-07-2026	13-07-2026
4	Firmware integration: Establish micro-ROS node on Pico and verify raw data publishing.	14-07-2026	18-07-2026
5	Central network setup: Configure ROS master laptop and wireless communications.	19-07-2026	23-07-2026
6	Vision tracking setup: Calibrate overhead camera and visual tracking sigil.	24-07-2026	28-07-2026
7	Control integration: Combine wheel encoder inputs, IMU readings, and camera frames.	29-07-2026	02-08-2026
8	PID parameter tuning: Calibrate controller gains for trajectory tracking stability.	03-08-2026	06-08-2026
9	Benchmarking execution: Run the seven performance metrics on the platform and TurtleBot.	07-08-2026	16-08-2026
10	Statistical evaluation: Analyze RMSE, latency, standard deviations, and generate comparative plots.	17-08-2026	22-08-2026
11	Report compiling: Draft final report, edit analysis sections, and incorporate advisor feedback.	23-08-2026	30-08-2026

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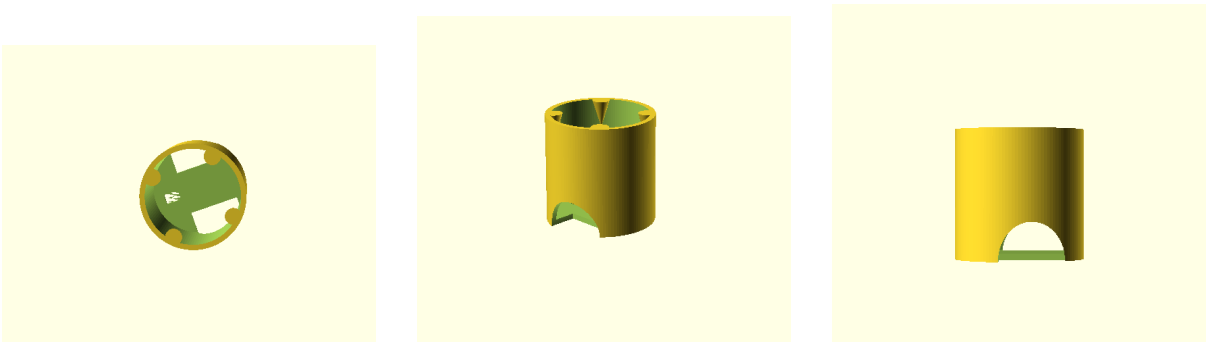
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Appendix A

3D CAD Models of the Robot Chassis

This appendix contains the three-dimensional computer-aided design (CAD) model renderings of the robot chassis body. Figure A.1 shows the top view, front view, and side view of the design.



(a) Top view. (b) Front view. (c) Side view.

Figure A.1: Three-dimensional CAD models of the robot chassis body.

Appendix B

Microcontroller Specification

This appendix presents the technical layout of the microcontroller. Figure B.1 illustrates the Raspberry Pi Pico board layout retrieved from the official documentation [Raspberry Pi Foundation 2026].

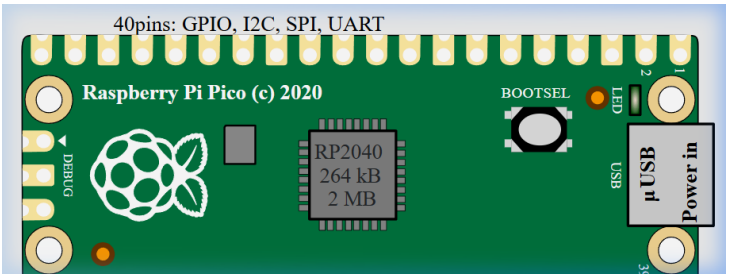


Figure B.1: Raspberry Pi Pico board layout [Raspberry Pi Foundation 2026].

Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☒
2. Sourcing Related Work (summarising, identifying sources) ☒
3. Methods and Experiment Design (experiment setup, model tuning) ☐
4. Data Analysis (presentation, coding, interpretation) ☐
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☐
7. Presentation (rendering graphics, formatting) ☐
8. Editing (grammar, readability) ☒
9. Writing (text generation, document structuring) ☒
10. Citation Formatting (structuring, organising) ☐

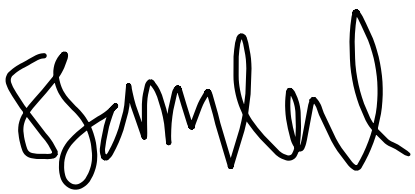
If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

Student number: 2459211

Candidate signature:



Date: 09-04-2026